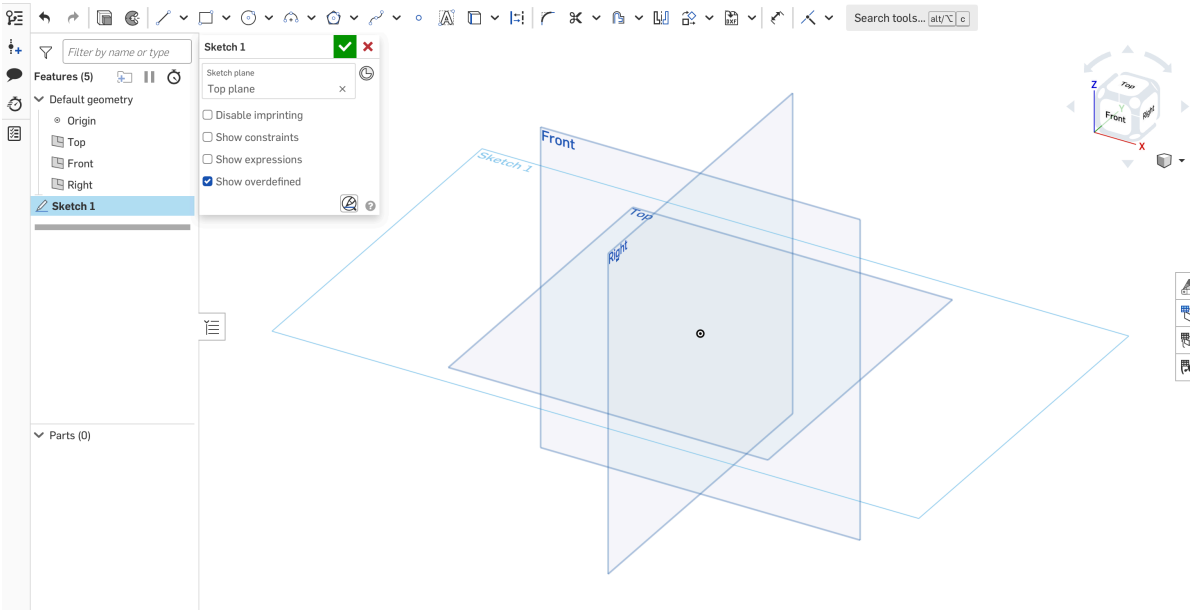
How to Convert an Image to an Onshape Sketch

A step-by-step guide to converting an image into a DXF sketch for CAD software



Step-by-Step Instructions

STEP 1 Upscale the Image

- Upload your image to iLoveIMG's upscaler.
- Select the highest upscale factor available (typically 2× or 4×).
- Download the upscaled PNG — this dramatically improves tracing quality in Step 2.

https://www.iloveimg.com/upscale-image

STEP 2 Convert the Image to SVG

- Go to FreeConvert's SVG converter.
- Upload your upscaled image and select SVG as the output format.
- Adjust the colour mode to Black & White if you only need outlines.
- Download the SVG file.

https://www.freeconvert.com/svg-converter

STEP 3 Convert SVG to DXF

- Upload your SVG to CloudConvert's SVG-to-DXF converter.
- Leave the settings at their defaults — Onshape is compatible with standard DXF R14+.
- Download the DXF file to your computer.

🔗 https://cloudconvert.com/svg-to-dxf

STEP 4 Import the DXF into Onshape

- Open your Onshape document and enter a Part Studio.
- Create a new Sketch on the plane you want the image on.
- Click Insert → DXF/DWG and upload your DXF file.
- Position and scale the imported geometry as needed.
- Close the sketch — the traced shape is now fully editable as Onshape sketch entities.

Troubleshooting

SVG has too many stray lines	Increase image contrast before upscaling, or use Inkscape to manually delete unwanted path segments.
DXF imports at wrong scale in Onshape	Use Sketch → Transform → Scale to resize. Set a known reference dimension to get the scale right.
Onshape won't accept the DXF	Re-export from CloudConvert choosing DXF version R14 or R2000. Avoid R2010+ as compatibility varies.
Tracing loses fine detail	Try a dedicated vectoriser such as Vectorizer.AI (free tier available) instead of FreeConvert for complex shapes.